

**JUSTIFICATION AND APPROVAL
FOR OTHER THAN FULL AND OPEN COMPETITION
SPRDL1-23-R-0145**

- 1. Contracting Agency:** Defense Logistics Agency - Land Warren (DLA-WRN).
- 2. Description of Action:** DLA Land Warren proposes to award a new Firm-Fixed-Price contract to Beaver Aerospace & Defense, Inc., Commercial and Government Entity (CAGE) 088K1, for National Stock Number (NSN) 6105-01-136-0964, Motor, Ball Screw, using Army Working Capital Funds.
- 3. Description of Supplies/Services:** The NSN 6105-01-136-0964, part number J-16116-20 / CAGE 088K1, Motor, Ball Screw, is used on the M860A1 Patriot Trailer Actuator. The deliverable for this contract is for a base quantity of 160 each, with an option quantity of 160 each, for a combined total quantity of 320 each. The total maximum value of this contract is \$ [REDACTED]

Acquisition Method Code (AMC):

3 - Acquire, for the second or subsequent time, directly from the actual manufacturer.

Acquisition Method Suffix Code (AMSC):

R - The Government does not own the data or the rights to the data needed to purchase this part from additional sources. It has been determined to be uneconomical to buy the data or rights to the data. It is uneconomical to reverse engineer the part. This code is used when the Government did not initially purchase the data and/or rights. If only one source has the rights or data to manufacture this item, AMCs 3, 4, or 5 are valid. If two or more sources have the rights or data to manufacture this item, AMCs 1 or 2 are valid.

4. Authority Cited: The statutory authority permitting other than full and open competition for this effort is 10 United States Code (U.S.C.) 3204(a)1) as implemented by the Federal Acquisition Regulation (FAR) 6.302-1, Only One Responsible Source and No Other Supplies or Services Will Satisfy Agency Requirements.

5. Reason for Authority Cited: NSN 6105-01-136-0964 is an alternating current (AC) Motor used to actuate the ball screw in the outrigger assembly on the M860 Patriot Trailer. If the Patriot Trailer did not contain this motor, the outriggers would be unable to deploy. The entire purpose of the Patriot trailer is to transport and provide a stable base for the Patriot Missile system to fire off. Without functioning outriggers, the trailer would be unable to provide this function. NSN 6105-01-136-0964 has only been supplied by two companies since the NSN was originally provisioned back in 1982. The original source of supply was Beaver Precision Products, now Beaver Aerospace and Defense. There was a secondary source added in the mid 80's, Sundstrand Corp, which has since been involved in multiple mergers and no longer exists. Since 1990, Beaver Aerospace

and Defense (CAGE 088K1), has been the sole source for this part. The only way to increase competition for this NSN, would be to reverse engineer the part and then search for a source to manufacture a 1G part, or for a company to willingly submit a Source Approval Request to become an approved source. Both of which would take an excessive amount of time and funding, resulting in a delay to alleviate current backorders for this part.

The Government does not have the pertinent technical data to enable competitive re-procurement. The Government does not own the necessary intellectual property data rights to the pertinent technical data to enable competitive re-procurement.

The Government does not have rights to the Technical Data Package (TDP) or specifications. Beaver Aerospace and Defense (CAGE 08K11) is the sole owner of the technical data package and the data rights are proprietary to them.

Beaver Aerospace and Defense (CAGE 08K11) is an industry leader in custom screw balls and aircraft actuation systems and uses advanced manufacturing processes to ensure the products can withstand the high pressure, extreme temperatures, and vibration of harsh aerospace environments. This specific ball screw motor is used on the M860 Patriot Missile System Outriggers and acts as the driving motor to deploy the outriggers. The Patriot Missile System is a critical system in the Army portfolio, and without this motor, the Patriot Trailer would be unable to perform its sole mission; to assist with the operation and act as a stable base for the Patriot Missile System when under operation. During the design of the M860 Patriot Trailer, Beaver Aerospace and Defense specifically developed this completely motorized ball screw package in order to level and stabilize the launch platform. Because of this, rigorous developmental testing tailored specifically to the mission of the M860, as well as functional testing during the Government's acquisition process has been conducted on this motor. This part has been sole sourced through Beaver Aerospace and Defense for the last 33 years. If the Army were to find another source for the part, there would not be that same level of confidence in the new source that has been built with Beaver Aerospace and Defense over the years through lab and functional testing, and successful missions completed by the Patriot System. This part is critical for the trailer to remain operational and there is a higher chance of a failure occurring when going with a new source who isn't familiar with the parts usage and the trailer itself.

Since there have been no vendors to come forward on their own expressing interest in becoming a supplier for this AC Motor, this leaves the Government with little options to increase competition. The first option would be to reach out to Beaver Aerospace and Defense to see if they would be interested in selling the technical data for this AC Motor, giving the Government the right to use the TDP for open and competitive procurement. Seeing as this is proprietary data from Beaver, and they are the sole source of this part, and still are in business of supplying this part, it is highly unlikely that they would be willing to sell the rights of the TDP to the Government.

The next option, would be for the Government to reverse engineer the part, create level III part drawings/TDP, and then using SAM.gov to give companies the chance

to bid on supplying this part. Although this would create an open and competitive environment for this part, this comes at a high cost to the Government, as it would entail procuring sample parts and getting them shipped to the Government facility, lab testing the Original Equipment Manufacturer (OEM) sample, reverse engineering and creating drawings for the part, creating prototypes, and then testing those prototypes in both the lab and on a trailer to make sure it's an identical replacement to the original OEM supplied part, in regards to form, fit, and function.

Leveraging past Reverse Engineering projects completed by Ground Vehicle System Center (GVSC) Prototype Integration Facility and test events led by Program Manager Heavy Tactical Vehicles, the estimated cost for completing the reverse engineering effort is laid out below.

Reverse Engineering Activity	Cost (\$)
Purchase of two AC motors/Shipping (six months)	
Estimate of Reverse Engineering/Drawing Creation (nine months)	
Lab Testing of Original Part (two months)	
Manufacture three (three) prototypes (four months)	
Lab Testing of Prototype parts (two months)	
Functional Testing of Installed Part (two months)	
TDP Finalization/Release (one month)	
Total Reverse Engineering Cost (25 months)	\$523,100

The Government would not be able to recover, through subsequent competition, the duplication of costs involved in reverse engineering the item to create a new and competitive TDP within a reasonable amount of time. At the current average usage rate of units each year, a current unit cost of \$, and assuming % cost savings achieved through competition, the \$ in annual savings would take just over seven years to offset the estimated cost of reverse engineering the item and developing a competitive TDP.

With the payback period required to recoup the cost of reverse engineer the part, along with the 24-month delay to complete the reverse engineering effort, it is not in the Government's best interest to delay the procurement of NSN 6105-01-136-0964 as there is currently a backorder quantity of each. With an annual demand of each, by the time the reverse engineering effort is completed to create an open and competitive environment, it is projected that the backorder quantities would increase to each. Since the Patriot Trailer would be dead lined due to a non-operational outrigger motor, this delay is unacceptable with a potential trailers being deemed non-mission capable before the sourcing process has even started, and would have a significant impact on the Army's ability to sustain the M860 Patriot Trailer fleet.

6. Efforts to Obtain Competition:

- a. Effective competition: Due to the reasons cited in section 5 above, the Government is not pursuing competition for this item.
- b. Sole Source: Notices required by FAR 5.201 will be posted.
- c. Subcontracting competition: Unless the contract is awarded to a small business, the solicitation and resulting contract will include FAR clauses 52.219-8, "Utilization of Small Business Concerns"; 52.244-5, "Competition in Subcontracting"; 52.219-16, "Liquidated Damages--Subcontracting Plan"; and Defense Federal Acquisition Regulation Supplement (DFARS) 252.219-7003, "Small Business Subcontracting Plan (DoD Contracts)," to maximize small business opportunities and utilization.

7. Actions to Increase Competition: There are no plans to increase competition for future procurements due to the reasons cited above.

There are no contractors currently in the process of being qualified as an alternate source of supply for this item. There are no current or planned efforts to qualify additional sources since the Government does not possess a fully defined and validated TDP for this item.

8. Market Research: A Sources Sought Notice was posted to SAM.gov on 15 June 2023, and was open for 15 days. No responses were received.

An internet search was performed June 2023 by GVSC Sustainment Engineering Team to see if there were like items offered by other manufacturers that may fit the Army's need. This ball screw motor was designed specifically to fit the mission of the M860 Patriot Trailer, which makes it very hard to find a commercial like-item that will satisfy the same form, fit, and function. Due to the lack of technical characteristics available to the Government, and the multitude of performance requirements categorizing ball screw motors, the validity of the researched commercial motor ball screws is difficult to evaluate. Results are below.

Keyword: Ball Screw AC Motors

Website: <https://www.helixlinear.com/Products/Stepper-Actuators-External-/Ball-Screw-Stepper-Actuators/412>

Part Numbers Looked at: BSA-34S2.85-E-B16020-FG-A-3.00-00-E200-A
BSA-34S2.85-E-B16020-FG-A-3.00-00-E200-A

Helix Linear sells Ball Screw AC Motors up to National Electrical Manufacturers Association (NEMA) 34 (example part numbers above), which is much smaller than the AC motor Beaver Aerospace and Defense is supplying for the Patriot Trailer outriggers. The Government does not have any of the technical specifications of the current part number beyond the physical size of the motor to even find a form, fit and function

replacement.

Website: <https://www.moonsindustries.com/c/ball-screw-hybrid-linear-stepper-motors-a0901004>

Moon Industries also sells many different Ball Screw AC Motors but does not sell anything beyond NEMA 23. These motors are also much smaller than the current part number being utilized by the Government.

Part Number's looked at: BE238S-B1002-100-AK1-0-220
BE238S-B1002-125-AK1-0-220
BE238S-B1002-150-AK1-0-220

Although these manufacturers supply many different sized ball screws, the Government was unable to identify one that met the envelope size of the current motor being used on the Patriot Trailer. Beyond the actual size of the motor, there would be endless technical requirements that would have to match in order to be considered a form, fit and function replacement. These would include, rated current (A), ball screw diameter and lead, rated thrust (N), linear travel per step, etc.

The Government also contacted the contractor to request consideration to purchase the intellectual property. No response was received. Beaver Aerospace and Defense has never expressed interest in selling the technical data for any part in the past.

9. Interested Sources: To date no additional sources have expressed interest in this item.

10. Other Facts: Procurement History:

Contract Number: W56HZV04D0005
Award Date: 30 October 2003
Contractor: Beaver Aerospace & Defense Inc.
Contract Value: \$1,194,216
Competitive Status: FAR 6.302-1 only one responsible source

11. Technical Certification: I certify that the supporting data under my cognizance which is included in the justification is accurate and complete to the best of my knowledge and belief.

Name: [REDACTED]
Title: Division Chief
Email: [REDACTED]@army.mil

Signature:

8/28/2023

X [REDACTED] 8/28/23

Signed by: [REDACTED]

12. Requirement Certification: I certify that the supporting data under my cognizance which is included in the justification is accurate and complete to the best of my knowledge and belief.

Name: [REDACTED]

Title: Division Chief

Email: [REDACTED]@army.mil

Signature:

8/28/2023

X [REDACTED]

Signed by: [REDACTED]

13. Fair and Reasonable Cost Determination: I hereby determine that the anticipated cost to the Government for this contract action will be fair and reasonable. This determination will be based on the results of a detailed cost and or price analysis. The contractor will not be required to submit certified cost and pricing data as this item has been deemed commercial.

Name: Reinbolt, Adam R CIV DLA LAND AND MARITIME (USA)

Title: Contracting Officer

Email: adam.reinbolt@dla.mil

Signature:

8/28/2023

X Adam Reinbolt

Signed by: REINBOLT.ADAM.ROSS.1294337225

14. Contracting Officer Certification: I certify that this justification is accurate and complete to the best of my knowledge and belief.

Name: Reinbolt, Adam R CIV DLA LAND AND MARITIME (USA)

Title: Contracting Officer

Email: adam.reinbolt@dla.mil

Signature:

8/28/2023

X Adam Reinbolt

Signed by: REINBOLT.ADAM.ROSS.1294337225

**Approval Authority for Acquisitions Equal to or Greater than \$750K But
Less Than \$15M**

APPROVAL OF JUSTIFICATION FOR
OTHER THAN FULL AND OPEN COMPETITION

Based on the forgoing Justification, I hereby approve the following procurement of the following:

- a. NOUN: Motor Ball Screw
- b. NSN: 6105-01-136-0964
- c. Basic Quantity: 160 Each
- d. Basic Estimated Value: [REDACTED]
- e. Option Percentage: 100%
- f. Quantity with Option: 320 Each
- g. Total Estimated Maximum Value with Option:
\$ [REDACTED]

This action will be awarded under other than full and open competition pursuant under the authority of 10 U.S.C. 3204(a)(1) and subject to the availability of funds. This approval is subject to the provision that the property herein described has otherwise been authorized for acquisition.

Signature:

✕ DAPO.DOUGLAS.A.123022 Digitally signed by
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Date: 2023.09.21 12:49:35 -04'00'